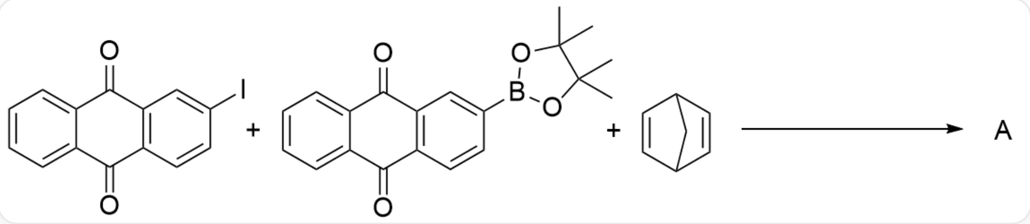
Question

Compound **A** was prepared from reactants of the following structures under Suzuki reaction conditions:



O=C1C2=C(C=CC(I)=C2)C(C3=CC=CC=C31)=O.O=C4C5=C(C=CC(B6OC(C)(C)C(C)(C)O6)=C5)C(C7=CC=CC=C74)=O.C89C=CC(C9)C=C8>>[*]

A can be used as a monomer to polymerize into polymers for battery manufacturing. During the polymerization process, one carbon-carbon bond in the monomer is completely broken.

Regarding **A** and the polymer obtained by polymerizing using **A** as a monomer, there are the following conclusions. Select the option corresponding to all incorrect conclusions.

Conclusion 1: When this polymer is directly used as a cathode material to make a battery, it can be used without charging.

Conclusion 2: 1 mol of this polymer formed by monomer polymerization requires the transfer of 4 mol of electrons for each reversible and complete charging or discharging process.

Conclusion 3: The polymerization reaction described can be catalyzed by **A** in Ziegler-Natta catalyst catalysis.

Conclusion 4: If the structural fragment connecting the aromatic ring units in the monomer is replaced with a cyclohexene structural fragment, the polymerization reaction will be more likely to occur.

Conclusion 5: If used in an aqueous electrolyte battery, this polymer is less likely to fall off the electrode during swelling than poly(2-vinyl)anthraquinone.

Conclusion 6: If only the main chain of the polymer is considered, the degree of unsaturation is 0.

Conclusion 7: This polymer is obtained by polymerization of $\mathbf{A}$ via olefin metathesis reaction.

Conclusion 8: This polymer is a linear polymer.

A. The following options do not meet the requirements.

B. Conclusion 1, Conclusion 4

C. Conclusion 2, Conclusion 3, Conclusion 4

D. Conclusion 1, Conclusion 3, Conclusion 6

E. Conclusion 2, Conclusion 4, Conclusion 5

F. Conclusion 2, Conclusion 5, Conclusion 6

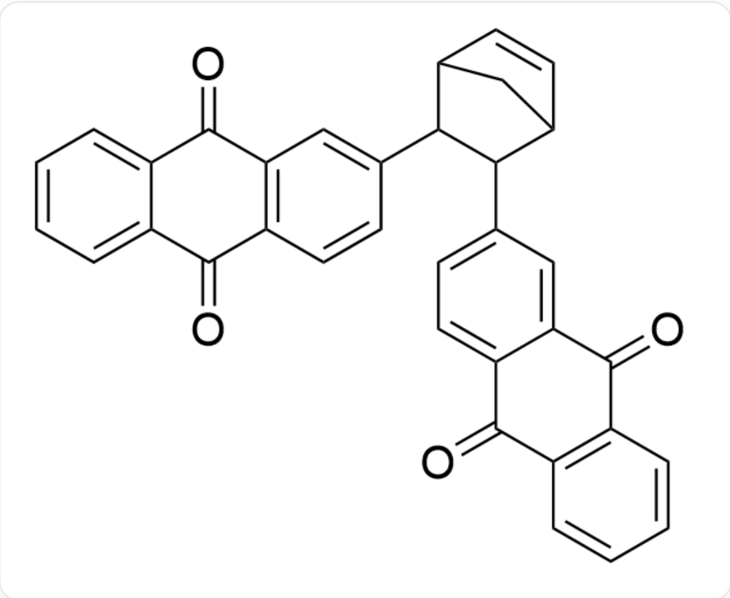
G. Conclusion 3, Conclusion 4, Conclusion 6

Answer

Correct Answer: G

Detailed Explanation

Based on the Suzuki reaction conditions, it can be inferred that the structure of **A** is



O=C(C(C=CC=C1)=C1C2=O)C(C2=C3)=CC=C3C(C4C=CC5C4)C5C6=CC7=C(C=C6)C(C8=CC=CC=C8C7=O)=O

If no other monomers are introduced, the structural fragment in the structure of **A** that can be used for polymerization reaction is only the norbornene fragment. The norbornene structure has a large ring strain and can undergo ring-opening polymerization through olefin metathesis reaction. Therefore, it can be inferred that the polymerization reaction of **A** is a ring-opening polymerization reaction that opens the norbornene structure.

CHECKPOINT 2 PTS

The polymerization reaction of **A** is a ring-opening polymerization reaction that opens the norbornene structure

Conclusion 1: A reduction reaction occurs at the cathode, and the anthraquinone fragment itself has oxidizing properties and can gain electrons to undergo a reduction reaction. Therefore, the battery can be used directly as a cathode material without charging. Conclusion 1 is correct.

CHECKPOINT 1 PTS

The anthraquinone fragment itself has oxidizing properties and can gain electrons to undergo a reduction reaction. Therefore, the battery can be used directly as a cathode material without charging

Conclusion 2: The reversible redox site in each monomer molecule is the anthraquinone fragment. Each anthraquinone fragment requires 2 electrons to be reversibly reduced. Each monomer contains 2 anthraquinone fragments, so each monomer requires 4 electrons to be reduced. Therefore, the polymer generated from 1 mol of monomer needs to transfer 4 mol of electrons during charging or discharging. Conclusion 2 is correct.

CHECKPOINT 1 PTS

Each monomer contains 2 anthraquinone fragments, so each monomer requires 4 electrons to be reduced

Conclusion 3: Ziegler-Natta catalysts are used to catalyze the coordination polymerization of olefins. During the polymerization process, only the π bond of the carbon-carbon double bond is broken, and the σ bond is not broken, which does not meet the requirement of the description "a carbon-carbon bond is completely broken". Conclusion 3 is incorrect.

CHECKPOINT 1 PTS

Ziegler-Natta catalysts are used to catalyze the coordination polymerization of olefins. During the polymerization process, only the π bond of the carbon-carbon double bond is broken, and the σ bond is not broken

Conclusion 4: The driving force for the ring-opening polymerization of **A** is the release of ring strain of the norbornene structure during the ring-opening process. If the norbornene fragment is replaced with a cyclohexene fragment, the ring strain will be greatly reduced, which is not conducive to ring-opening polymerization. Conclusion 4 is incorrect.

CHECKPOINT 1 PTS

Replacing the norbornene fragment with a cyclohexene fragment will greatly reduce the ring strain, which is not conducive to ring-opening polymerization

Conclusion 5: During the swelling process, the anthraquinone anion group has strong hydrophilicity, which easily leads to the detachment of the electrode material, while the polynorbornene long chain is more hydrophobic than the polyethylene long chain, thus stabilizing the electrode material. Conclusion 5 is correct.

CHECKPOINT 1 PTS

The polynorbornene long chain is more hydrophobic than the polyethylene long chain

Conclusion 6: The main chain of the polymer contains unsaturated double bonds. Conclusion 6 is incorrect.

CHECKPOINT 1 PTS

The main chain of the polymer contains unsaturated double bonds.

Conclusion 7: The polymerization reaction of **A** is a ring-opening polymerization reaction that opens the norbornene structure. Conclusion 7 is correct.

Conclusion 8: The polymerization reaction of **A** is a ring-opening polymerization reaction that opens the norbornene structure and will not produce any side chains, so it is a linear polymer. Conclusion 8 is correct.